Over the past few centuries, physicists have sought to consolidate our understanding of the natural world by showing that diverse and apparently distinct phenomena are actually governed by a single set of physical laws. To Einstein, this goal of unification—of explaining the widest array of phenomena with the fewest physical principles—became a lifelong passion. With his two theories of relativity, Einstein united space, time, and gravity. But this success only encouraged him to think bigger. He dreamed of finding a single, all-encompassing framework capable of embracing all of nature's laws; he called that framework a *unified theory.* Although now and then rumors spread that Einstein had found a unified theory, all such claims turned out to be baseless; Einstein's dream went unfulfilled.

Einstein's focus on a unified theory during the last thirty years of his life distanced him from mainstream physics. Many younger scientists viewed his single-minded search for the grandest of all theories as the ravings of a great man who, in his later years, had turned down the wrong path. But in the decades since Einstein's passing, a growing number of physicists have taken up his unfinished quest. Today, developing a unified theory ranks among the most important problems in theoretical physics.

For many years, physicists found that the central obstacle to realizing a unified theory was the fundamental conflict between the two major breakthroughs of twentieth-century physics: general relativity and quantum mechanics. Although these two frameworks are typically applied in vastly different realms—general relativity to big things like stars and galaxies, quantum mechanics to small things like molecules and atoms—each theory claims to be universal, to work in all realms. However, as mentioned above, whenever the theories are used in conjunction, their combined equations produce nonsensical answers. For instance, when quantum mechanics is used with general relativity to calculate the probability that some process or other involving gravity will take place, the answer that's often found is not something like a probability of 24 percent or 63 percent or 91 percent; instead, out of the combined mathematics pops an *infinite* probability. That doesn't mean a probability so high that you should put all your money on it because it's a shoo-in. Probabilities bigger than 100 percent are meaningless. Calculations that produce an infinite probability simply show that the